

Realization Limits of the Bicomplex Signal Surface: Analytic Rigidity, Observation Monodromy and Corrections

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Working draft of the corrected successor to *Geometric Realization of Bicomplex Signal Manifolds*
(v11/v12)

Abstract

Earlier versions of this work asserted that a moment map and a power constraint, followed by a “geometric projection” $\mathbb{R}^4 \rightarrow \mathbb{R}^3$, produce an explicit ruled surface S . We prove that the historical moment assignment $(|z_1|^2, |z_2|^2) = (\cos t, 1 - \cos t)$ admits no readout that is real-analytic along the circle $\{z_1 = 0\}$, whatever family of states is used. This excludes affine projections, real polynomial maps and every polynomial in a bicomplex variable and its conjugates. Phase-invariant and mixed-state readouts fail without any regularity hypothesis. The obstruction is sharp in three ways. An explicit continuous state family with an explicit C^∞ readout realizes S . Each half of the source admits a real-analytic readout but no affine one. A different power assignment admits a phase-sensitive quadratic readout.

We then separate the carriers of the observation-field monodromy. The square-root cover gives a rank-two local system that is indecomposable over $\mathbb{Z}$ and splits only after inverting 2. It differs from the Whitney fold involution and from the normalization defect, and the constant sheaf carries no ± 1 decomposition. For every continuous map from a convex parameter domain to the plane, the exceptional pullback of $\mathbb{Z}$ is the extension by zero from the interior. Folds are detected instead by failure of the comparison map that defines cohomological smoothness. Finally, we record the residual geometry and spectral facts that survive, with their exact scope. An appendix corrects all 66 named statements of the earlier version, one statement specific to its first public file, and the earlier public description, item by item.

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1 Introduction

The preprint *Geometric Realization of Bicomplex Signal Manifolds* has two public files, v11 and v12 [1]. It studied an explicit ruled map S , together with a signal-theoretic narrative: a power-conserving two-component state, a moment map, a projection to three-dimensional observations, and a family of spectral, sheaf-theoretic and arithmetic interpretations. A complete audit of the 66 named statements of v12, together with new proofs, shows that the geometry of S is rich and largely correct. Several of the interpretive bridges, however, are either unsupported or false. This paper replaces that record. Its own new results concern the step the title of the earlier work promised and never supplied: *which readouts of which state families realize S ?*

Main results.

1. *Analytic rigidity* (Theorem 3.6). Under the historical assignment $|z_1|^2 = \cos t$, every state over the two edges $t = \pm\pi/2$ lies on the circle $C_0 = \{z_1 = 0\}$, and those edges map to a non-collinear “V”. Consequently no readout that is real-analytic along C_0 realizes S , for any state family, on the closed source or on the source open in t .
2. *Sharpness*. An explicit C^∞ readout realizes S (Theorem 3.8). On one half of the source, real-analytic readouts exist but affine ones do not (Proposition 3.9). A different power assignment admits a phase-sensitive quadratic readout (Theorem 3.2). Phase-invariant readouts and readouts of mixed states fail outright (Proposition 3.4). Remark 3.10 gives an edge criterion that explains all of these.
3. *Carriers of the monodromy* (Proposition 4.3). The two-traversal restoration of $\sqrt{F}$ lives on a local system that is indecomposable over $\mathbb{Z}$. It is distinct from the fold involution and from the normalization defect.
4. *Exceptional pullbacks* (Theorem 4.4 and Proposition 4.6). $\Psi^!\mathbb{Z}$ depends only on the convex parameter domain, not on the map; the earlier fold formula $i^*\Psi^!\mathbb{Z} \simeq \mathbb{Z}[-1]$ holds at no point.

Sections 5–7 retain the correct residual geometry, spectral estimates and scope limits of v12, in sharpened form. Appendix A lists every correction.

What “bicomplex” means here. The bicomplex algebra is isomorphic, through its idempotents, to $\mathbb{C} \oplus \mathbb{C}$. Its three standard conjugations become componentwise conjugation and the exchange of the two components. No proof below uses bicomplex multiplication beyond this identification. That is itself one of the outcomes: readouts built from bicomplex multiplication and conjugations are real polynomial maps, so Corollary 3.7 excludes them as well. We keep the name for continuity with the earlier record.

Relation to other work. The completed source atlas, the five rank singularities, the dipole indices and the fold classification of the planar observation map are proved for the same surface in the published papers [2, 3]. We cite them rather than re-prove them. The scalar Green, Markov and Krein theory of the Laplace–Beltrami operator, and the corrected two-point extension problem that replaces v12 Propositions 8.13–8.16, belong to a separate companion paper [4]. Standard frameworks are credited where used. The new contributions are the model results named above. Their priority relative to the literature has not been systematically established.

Evidence. Every statement carries one of the following kinds of evidence: a written proof; a cited external theorem with its hypotheses checked; an exact computer-algebra identity; a finite numerical replay; or a Lean-checked statement. Appendix B states which applies to which result. No result rests on a historical “proven” label.

2 The surface, its source and the observation field

Throughout,

$$S(t, u) = (c + 2u(1 - c), (1 - u)s, uQ), \quad c = \cos t, \quad s = \sin t, \quad Q = \sqrt{c(2 - c)}, \quad (1)$$

on the source $D = [-\pi/2, \pi/2] \times [0, 1]$. Put $D_\circ = (-\pi/2, \pi/2) \times [0, 1]$ and $D_+ = [0, \pi/2] \times [0, 1]$. The edges and the polar ruling map to

$$S(\pm \frac{\pi}{2}, u) = (2u, \pm(1 - u), 0), \quad S(0, u) = (1, 0, u). \quad (2)$$

The two edge segments lie on distinct lines through the apex $(2, 0, 0)$. The ruling line $\{(1, 0, z)\}$ is skew to both edge lines.

Proposition 2.1 (Completed source and five rank labels [2]). *Near $t = \epsilon\pi/2$, $\epsilon \in \{1, -1\}$, let $r = Q \geq 0$ and $C(r) = 1 - \sqrt{1 - r^2}$, and put $B_\epsilon(r, u) = (C + 2u(1 - C), \epsilon(1 - u)\sqrt{1 - C^2}, ur)$. This extends smoothly to $|r| < 1$ and agrees with S on the physical overlap. The rank-loss labels of the completed source are exactly*

$$P_0 = (0, 1), \quad P_\pm = (\pm t_0, u_0), \quad E_\pm = (\pm \pi/2, 0), \quad c_0 = \frac{3 - \sqrt{5}}{2}, \quad t_0 = \arccos c_0, \quad u_0 = \frac{1 - c_0}{2}.$$

Their smooth extensions are cross-caps. The adapted determinants are $\Delta(P_0) = 1$, $\Delta(P_\pm) = -5\sqrt{\sqrt{5} - 2}$ and $-\epsilon$ at $E_\pm$. The physical restrictions at P_0 and $E_\pm$ are a boundary half-germ and corner quarter-germs.

Sketch. $C(r)(2 - C(r)) = r^2$ gives $C(Q) = c$, and differentiation gives $(B_\epsilon)_r \times (B_\epsilon)_u = (\epsilon u, 2u, 0)$ at $r = 0$. On $c > 0$ the cross-product equations reduce to $c = 1, u = 1$, or to $u = (1 - c)/2 = c(2 - c)/2$, that is $c^2 - 3c + 1 = 0$. Since $S_{uu} = 0$, the cross-cap determinant reduces to $\det(S_u, S_{ut}, S_{tt})$, and the smooth recognition criterion [5, Cor. 4.5] applies. Full proofs and Lean-checked rank statements are in [2]. $\square$

Thus $c^2 - 3c + 1$ selects only the two lateral labels. On $D_{\text{reg}} = D \setminus \{P_0, P_\pm, E_\pm\}$, a connected orientable surface with corners, $H_1(D_{\text{reg}}; \mathbb{Z}) = \mathbb{Z}^2$.

The observation field. On the angular chart $|t| < \pi/2$ the normal $V = S_t \times S_u$ has components $V_z = N$ and $V_x = M$, where

$$N = (1 - c)(1 - c - 2u), \quad M = \frac{c^2(2 - c) - u(1 - c + c^2)}{Q}, \quad F = N + iM, \quad \Psi = (N, M). \quad (3)$$

Both components are even in t . Because $Q = 0$ at $t = \pm\pi/2$, M is unbounded there for $u > 0$. The field F and the map Ψ are therefore defined on $D_\circ$, not on D . They also depend on the chart: a reparametrization of the source multiplies V by its Jacobian.

3 Realizations of the surface by signal states

Let $S^3 = \{z \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}$, and let $C_0 = \{(0, w) : |w| = 1\}$ and $C_1 = \{(w, 0) : |w| = 1\}$. The moment map $z \mapsto (|z_1|^2, |z_2|^2)$ sends S^3 onto the segment $X + Y = 1$. By itself it defines neither a two-dimensional source, nor a lift of (1), nor a projection taking that lift to S . A realization therefore consists of two pieces of data.

Definition 3.1. Let $D' \subset D$. A *state family* on D' is any map $z : D' \rightarrow S^3$; no continuity is assumed. A *readout* is a map $P : U \rightarrow \mathbb{R}^3$ defined on a set $U \subset \mathbb{C}^2$ containing $z(D')$. The pair *realizes* S if $P \circ z = S$ on D' . The family is *moment-compatible* (with the historical assignment) if $|z_1(t, u)|^2 = \cos t$. A readout is *real-analytic along* C_0 if $C_0 \subset U$ and $\beta \mapsto P(0, e^{i\beta})$ is real-analytic.

3.1 A realization with varying powers

The following realization exists, but with powers that differ from the historical ones.

Theorem 3.2 (Quadrature realization). *Put $\zeta_+(t) = e^{it/2}$ and $\zeta_-(t) = \sqrt{1 - c/2} + i\sqrt{c/2}$. The state family and the fixed real quadratic readout*

$$\Phi(t, u) = (\sqrt{1 - u}\zeta_+(t), \sqrt{u}\zeta_-(t)), \quad P(z) = (\Re z_1^2 + |z_2|^2 + \Re z_2^2, \Im z_1^2, \Im z_2^2)$$

satisfy $|\Phi_1|^2 = 1 - u$, $|\Phi_2|^2 = u$ and $P \circ \Phi = S$ on all of D . The family is smooth for $0 < u < 1$ in the completed charts.

Proof. $\zeta_+^2 = c + is$ and $\zeta_-^2 = 1 - c + iQ$, so $P\Phi = ((1 - u)c + u(2 - c), (1 - u)s, uQ) = S$. In the edge chart the imaginary part of ζ_- is $r/\sqrt{2(2 - C)}$ and its real part is $\sqrt{1 - C/2}$; both are smooth. $\square$

The required phase coupling is model data; power conservation does not select it.

Proposition 3.3 (Two obstructions). *(1) No C^1 state family can have $|z_2|^2 = u$ up to $u = 0$, or $|z_1|^2 = 1 - u$ up to $u = 1$. (2) No three fixed Hermitian 2×2 matrices H_j and normalized pure-state family satisfy $S_j = z^* H_j z$ on D .*

Proof. (1) At a zero of a C^1 component w , the derivative of $|w|^2$ is $2\Re(\bar{w}\partial w) = 0$, while $\partial_u u = 1$. (2) $z^* H_j z = a_j + (Mr(z))_j$, where $r(z)$ is the Bloch vector, a point of the unit sphere, and M is a real 3×3 matrix. If $\text{rank } M \leq 2$ the image is planar, but the values $(1, 0, 0), (0, 1, 0), (0, -1, 0), (1, 0, 1)$ of S have affine determinant 2. If M is invertible the image is an ellipsoid, which contains no segment, but $S(0, u) = (1, 0, u)$ is one. See also [6]. $\square$

3.2 The historical assignment

Every state with $t = \pm\pi/2$ lies on C_0 , and every state with $t = 0$ lies on C_1 .

Proposition 3.4 (Phase-invariant and mixed-state readouts). *Suppose D' contains two distinct points of one edge $\{\pm\pi/2\} \times [0, 1]$. Then no readout with $P(e^{i\gamma}z) = P(z)$ realizes S on D' with a moment-compatible family. In particular no function of zz^* does, including all Hermitian expectation values. If trace-one density matrices $\rho \geq 0$ with $\rho_{11} = \cos t$ replace pure states, no function of ρ realizes S .*

Proof. C_0 is a single orbit of the global phase, so such a readout is constant on C_0 , whereas S takes distinct values at the two edge points. For density matrices, $\rho_{11} = 0$ and positivity force $\rho_{12} = 0$, so $\rho = \text{diag}(0, 1)$. $\square$

Lemma 3.5 (Analytic curves and lines). *Let $p : J \rightarrow \mathbb{R}^3$ be real-analytic on an open interval or on the circle. For every line ℓ , either $p(J) \subset \ell$ or $p(J) \cap \ell$ is countable. In particular $p(J)$ cannot contain uncountable subsets of two distinct lines.*

Proof. Choose $x_0 \in \ell$ and unit normals n_1, n_2 to ℓ . If $p(J) \not\subset \ell$, some $n_k \cdot (p - x_0)$ is a nonzero real-analytic function. Its zero set is discrete, hence countable, and contains $p^{-1}(\ell)$. Two distinct lines meet in at most one point. $\square$

Theorem 3.6 (Analytic rigidity). *Let a moment-compatible family z and a readout P realize S on D' .*

1. *If $D' = D$, then P is not real-analytic along C_0 . More generally, P is not real-analytic on any connected open arc of C_0 that contains all edge states $z(\pm\pi/2, u)$.*
2. *If $D' = D_\circ$, $C_0 \subset U$ and $P|_{U \cap S^3}$ is continuous at each point of C_0 , then P is not real-analytic along C_0 .*

Proof. (1) The image of C_0 , or of the arc, contains both edge segments of (2), which are uncountable subsets of distinct lines; this contradicts Lemma 3.5. (2) Fix u and $\epsilon = \pm 1$, and take $t_n \rightarrow \epsilon\pi/2$. A subsequence of $z(t_n, u) \in S^3$ converges to some z^* with $z_1^* = 0$, that is $z^* \in C_0$. Continuity gives $P(z^*) = \lim S(t_n, u) = S(\epsilon\pi/2, u)$, so both edge segments lie in $P(C_0)$. $\square$

Corollary 3.7 (Excluded projections). *Under the historical assignment on D or $D_\circ$, S is not realized by any affine map $\mathbb{R}^4 \rightarrow \mathbb{R}^3$, any real polynomial map, any polynomial in a bicomplex variable and its conjugates, or any map real-analytic on a neighbourhood of C_0 . On D this also excludes stereographic projection from a point of C_0 that avoids the edge states. In particular, the “Geometric Projection” of v12 Definition 3.1 cannot be completed by any such map.*

3.3 Sharpness

Theorem 3.8 (C^∞ realization). *Put $\theta = \pi/8$. Take smooth functions $\Lambda, \psi, \varrho, \kappa : [0, 1] \rightarrow [0, 1]$ such that:*

- $\Lambda = 0$ on $[0, \frac{1}{4}]$, $\Lambda = 1$ on $[\frac{3}{4}, 1]$, and Λ is strictly increasing in between;
- $\psi = 0$ on $[0, \frac{1}{3}]$ and $\psi = 1$ on $[\frac{2}{3}, 1]$;
- $\varrho = 0$ on $[0, \frac{3}{4}]$ and $\varrho = 1$ on $[\frac{7}{8}, 1]$;
- $\kappa = 1$ on $[0, \frac{1}{8}]$ and $\kappa = 0$ on $[\frac{1}{4}, 1]$.

Let χ be a smooth 2π -periodic function equal to 1 on $[-\frac{\pi}{8}, \frac{3\pi}{8}]$ and to -1 on $[\frac{7\pi}{8}, \frac{11\pi}{8}]$, and let $\nu = 0$ for $t \geq 0$ and $\nu = 1$ for $t < 0$. Define

$$z(t, u) = (\sqrt{c} e^{i\theta\Lambda(c)u}, \sqrt{1-c} e^{i(\nu\pi + \theta(1-\Lambda(c))(1+u))}).$$

This is a moment-compatible family. It is continuous on D , smooth on $D_\circ$, and smooth up to $t = \pm\pi/2$ in the edge charts, since $\sqrt{C(r)} = r/\sqrt{1 + \sqrt{1-r^2}}$. With $c = |z_1|^2$ and principal arguments, set

$$\begin{aligned} \mathcal{U} &= \psi(c) \frac{\arg z_1}{\theta\Lambda(c)} + (1 - \psi(c)) \left(\frac{\arg z_2^2}{2\theta(1 - \Lambda(c))} - 1 \right), \\ \Sigma &= \varrho(c) \Re z_2 \sqrt{1+c} + (1 - \varrho(c)) |z_2| \chi(\arg z_2) \sqrt{1+c}, \\ \mathcal{Q} &= \kappa(c) \Re z_1 \sqrt{2-c} + (1 - \kappa(c)) \sqrt{c(2-c)}, \end{aligned}$$

where each summand is read as zero where its cutoff vanishes. Then $P = (|z_1|^2 + 2\mathcal{U}|z_2|^2, (1 - \mathcal{U})\Sigma, \mathcal{U}\mathcal{Q})$ is smooth on an open neighbourhood of $z(D)$, and $P \circ z = S$ on D . Multiplying by a smooth bump equal to 1 near $z(D)$ gives a C^∞ readout on $\mathbb{C}^2$.

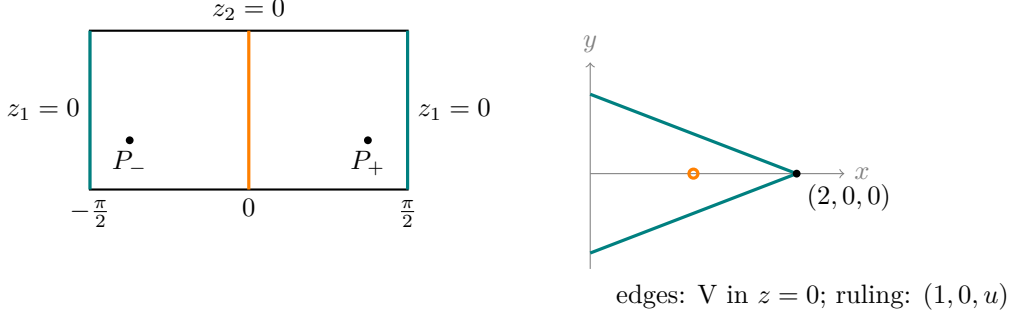


Figure 1: Under the historical powers, z_1 vanishes on both edges and z_2 on the polar ruling. The edges map to a V in the plane $z = 0$, which no analytic image of C_0 can contain. The ruling maps onto a segment over $(1, 0)$, on a line skew to both legs.

Proof. For $c \geq \frac{3}{4}$, $\Lambda = 1$ and $z_2 = \text{sgn}(t)\sqrt{1 - \cos t} = \sqrt{2}\sin(t/2)$, which is smooth; at $t = \pm\pi/2$ only $\sqrt{c} \rightarrow 0$ enters. On the image, $\arg z_1 = \theta\Lambda u \in [0, \theta]$, $\arg z_2^2 = 2\theta(1 - \Lambda)(1 + u) \in [0, \pi/2]$, and $\arg z_2 \in \nu\pi + [0, \pi/4]$. Hence both decoders equal u wherever they are used. The first is used only for $c \geq \frac{1}{3}$, where $\Lambda > 0$ and $z_1 \neq 0$. The second is used only for $c \leq \frac{2}{3}$, where $\Lambda < 1$ and $|z_2|^2 \geq \frac{1}{3}$.

Where $\varrho > 0$, z_2 is real and $\Re z_2\sqrt{1+c} = \text{sgn}(t)\sqrt{1-c^2} = s$. Where $\varrho < 1$, $\chi(\arg z_2) = \text{sgn } t$. Where $\kappa > 0$, $z_1 = \sqrt{c}$ and $\Re z_1\sqrt{2-c} = Q$. Each summand is smooth on the union of two open sets: one where its decoder is smooth, and one where its cutoff vanishes. The two definitions agree on the overlap, and the union contains the image. Substitution gives $P \circ z = S$; compactness of the image provides the bump. $\square$

A finite replay on 801×81 labels reproduces S to 1.4×10^{-14} . On C_0 the readout traces the two legs of the V on disjoint arcs, joined through the flat function χ . That is exactly what Lemma 3.5 forbids for analytic readouts.

Proposition 3.9 (One half of the source). *(1) If D' contains at least three points of the ruling $\{0\} \times [0, 1]$ and at least three points of one edge, then no affine readout realizes S on D' with a moment-compatible family. This applies in particular to D_+ , which contains the single interior cross-cap P_+ . (2) On D_+ there exist a real-analytic moment-compatible family and a readout that is real-analytic on a neighbourhood of its image and realizes S .*

Proof. (1) Write $P = b + L$. On C_1 the readout traces an ellipse centred at b , possibly degenerate, through three collinear points of the ruling line. A nondegenerate ellipse meets a line in at most two points, so the ellipse is a segment, and b lies on the ruling line. Applying the same argument on C_0 puts b on the edge line. The two lines are skew.

(2) Fix $0 < \theta < \pi$ and put $z = e^{i\theta u}v(t)$, with $v = (\sqrt{\cos t}, \sqrt{2}\sin(t/2))$. In the edge chart, $v = (r/\sqrt{1 + \sqrt{1 - r^2}}, (1 - r^2)^{1/4})$. Both expressions extend analytically across $t = 0$ and $r = 0$, and so does S in the completed charts. Since $\partial_u z = i\theta z$ and $\partial_t z = e^{i\theta u}v'$ with $v' \neq 0$ real, z is an immersion. If $e^{i\theta u}v = e^{i\theta u'}v'$ with v, v' nonnegative, the phase quotient is positive, so $u = u'$ and $t = t'$. An immersion that is injective on a compact set embeds a neighbourhood of it. The nearest-point retraction onto a real-analytic embedded surface is real-analytic on a tube, by the analytic inverse function theorem for $(x, n) \mapsto x + n$ on the normal bundle. Composing the retraction, the inverse embedding and the extended S gives the readout. $\square$

Remark 3.10 (Edge criterion and the extra data a realization needs). Lemma 3.5 yields a necessary criterion for any power assignment $w = |z_1|^2$. An analytic readout requires $S(w^{-1}(0))$ to lie in the image of an analytic loop, and a phase-invariant readout requires it to be a single point; the same holds for $1 - w$. The assignment of Theorem 3.2 passes. Its vanishing edges $u = 1$ and

$u = 0$ map to arcs of the circles $(x - 1)^2 + z^2 = 1$ and $x^2 + y^2 = 1$, and its readout maps the two vanishing circles onto exactly these circles. The historical assignment fails because $w^{-1}(0)$ consists of both edges.

A realization of the historical model therefore needs data the moment map does not supply. One option is a non-analytic switching readout, as in Theorem 3.8. Another is to keep only half of the source, which loses a cross-cap and the reflection pairing $S(t, 1) = S(-t, 1)$. A third is to change the powers. Approximation carries no content: by the Weierstrass theorem, polynomial readouts satisfy $\sup_D |P_\varepsilon \circ z - S| < \varepsilon$ for every $\varepsilon > 0$, yet none is exact. Whether a quadratic readout exists on D_+ is open.

4 Observation monodromy and its carriers

4.1 Interior dipole and square-root restoration

Theorem 4.1 (Dipole and two-traversal restoration). *The interior zeros of F in $D_\circ$ are exactly $P_\pm$, and $F(0, 1) = 0$ at the boundary. With the (t, u) orientation, $\det DF(P_\pm) = \mp 5s_0(1 - c_0)^2/Q_0$ and $\text{Ind}(P_\pm) = \mp 1$ [2]. Every continuous branch of $\sqrt{F}$ transported round a small positive loop about either point changes sign, and two traversals restore it.*

Proof. $N = 0$ exactly on $c = 1$ and on $u = (1 - c)/2$. On the first, $M = 1 - u$; on the second, M is a positive multiple of $-(c^2 - 3c + 1)$. The determinant follows from $c_0^2 = 3c_0 - 1$, and a nondegenerate zero has small-circle winding equal to the sign of the determinant. Write $F(\gamma(s)) = \exp L(s)$ with a continuous logarithm, so that $L(1) = L(0) + 2\pi i n$ with $n = \pm 1$. Then $\chi = \exp(L/2)$ satisfies $\chi(1) = -\chi(0)$. Any other continuous root differs from χ by a constant sign. $\square$

This is a count of traversals. It assigns no physical clock, and it does not identify the root cover with any spin structure. Earlier versions mixed this with two other involutions, which Proposition 4.3 separates. The value $|\det DF(P_\pm)| \approx 2.245$ is not $\sqrt{5}$. The number $\sqrt{5}$ is $|P'(c_0)|$ for the monic $P(c) = c^2 - 3c + 1$, and it changes when P is rescaled. Both N and M are even in t , so the sign reversal of the Jacobian comes from the t -derivatives, not from parity of the components.

4.2 The polar fold

Proposition 4.2 (Polar fold). *The smooth extension of Ψ at $P_0 = (0, 1)$ is a Whitney fold of local degree zero. No interior winding index is assigned on the physical half-neighbourhood.*

Proof. Use the source coordinates $a = -M$ and $b = 2 \sin(t/2) \sqrt{u - (1 - \cos t)/2}$, and the target coordinates $(N, M) \mapsto (-M, -N)$. The derivative matrix at $(0, 1)$ is invertible, and the exact identity $-N = b^2$ gives the normal form (a, b^2) . Reflecting b pairs preimages of opposite orientation. $\square$

The remainder displayed in v12 omitted $M(t, 1) = -t^4/2 + O(t^6)$, but the exact charts above do not depend on it. The full interior critical locus of Ψ consists of the symmetry segment $t = 0$, which is an ordinary fold for $u > 0$, and a rational branch, which consists of ordinary folds away from $(0, 0)$; at $(0, 0)$ the ordinary-fold condition fails [3, Thm. 2.4].

4.3 Three different sign structures

Proposition 4.3 (Carriers). *(1) For connected X , $\text{End}_{D(X)}(\mathbb{Z}_X) = H^0(X; \mathbb{Z}) = \mathbb{Z}$. Every involution of $\mathbb{Z}_X$ is therefore $\pm \text{id}$, $\mathbb{Z}_X$ admits no splitting into two nonzero summands, and its characteristic polynomial is $t - 1$ or $t + 1$. The same holds for the constant condensed sheaf on a compact connected space, whose global sections are the continuous maps to $\mathbb{Z}$.*

(2) Let $X = D_\circ \setminus \{P_+, P_-, (0, 1)\}$ and let $\pi : \tilde{X} \rightarrow X$ be the double cover $\{\chi^2 = F\}$. Then $\pi_*\mathbb{Z}$ is a rank-two local system with deck involution τ , indecomposable over $\mathbb{Z}$. After inverting 2, $\pi_*\mathbb{Z}[\frac{1}{2}] \simeq \mathbb{Z}[\frac{1}{2}] \oplus \mathcal{L}[\frac{1}{2}]$, where the sign system $\mathcal{L}$ has monodromy -1 around each of $P_\pm$.

(3) A Whitney fold involution $(\xi, \eta) \mapsto (\xi, -\eta)$ fixes the fold arc pointwise, whereas τ is fixed-point free; neither induces the other.

(4) For a standard cross-cap image X' with topological normalization ν , the rational normalization defect $\mathcal{Q}_\nu = \nu_*\mathbb{Q}_{\tilde{X}'}/\mathbb{Q}_{X'} \simeq j_!\mathcal{L}_{\text{sgn}}$ lives on the image double ray and has zero stalk at the pinch. Every ordinary morphism from $\mathcal{Q}_\nu$ to a skyscraper at the pinch is zero.

Proof. (1) Degree-zero endomorphisms of a sheaf in degree zero are the global sections of the sheaf $\mathcal{H}om(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}$. The only idempotents of $\mathbb{Z}$ are 0 and 1. (2) $D_\circ \setminus \{(0, 1)\}$ is star-shaped about an interior point, so $\pi_1(X)$ is free on loops around $P_\pm$, on which F winds ∓ 1 . The monodromy of π is therefore the surjection onto C_2 , and the stalk is the regular module $\mathbb{Z}[C_2] = \mathbb{Z}[x]/(x^2 - 1)$, whose only idempotents are 0 and 1. Over $\mathbb{Z}[\frac{1}{2}]$ the projectors $(1 \pm x)/2$ split it. (3) The fixed-point sets differ. (4) This is the stalk computation of v12 Proposition 8.6 [10]: the diagonal stalk map is an isomorphism at one-preimage points and has an anti-invariant one-dimensional quotient at two-preimage points. $\square$

Hence the “rank-two ± 1 decomposition of the constant sheaf” stated in the earlier public description has no carrier. The correct -1 sector is $\mathcal{L}$, which lives on the punctured source and records the winding of F . The normalization sign line lives instead on the image double stratum and records the exchange of sheets. The present data supply no canonical isomorphism between these two $\mathbb{Z}/2$ -torsors. No map from a Mellin Hilbert space to either carrier has been constructed.

4.4 Exceptional pullbacks do not see folds

We use the six-functor formalism on finite-dimensional locally compact Hausdorff spaces with $D(X) = D(\text{Sh}(X; \mathbb{Z}))$ [7, Ex. 5.9], $\omega_X = a_X^!\mathbb{Z}$ and $\mathbb{D}_X = \text{RHom}(-, \omega_X)$. Three standard facts are needed:

- Verdier duality, $f^!\mathbb{D}_Y \simeq \mathbb{D}_X f^*$ [8, Prop. 4.6.9], [9, Ch. III];
- the stalk of ω_X at x is the local homology $H_*(X, X \setminus x)$, and $\omega_X \simeq \text{or}_X[n]$ on an n -manifold [8, Rem. 4.6.19, Cor. 4.6.20];
- open immersions satisfy $j^! = j^*$ and are cohomologically smooth [7, Rem. 5.4].

A map f is cohomologically smooth when the comparison $f^!\mathbb{Z} \otimes f^*(-) \rightarrow f^!(-)$ is an isomorphism, $f^!\mathbb{Z}$ is invertible, and both properties survive base change [7, Def. 5.1].

Theorem 4.4 ($f^!\mathbb{Z}$ sees only dimension and boundary). (1) For any continuous $f : X \rightarrow Y$ between topological manifolds without boundary of dimensions n and m , $f^!\mathbb{Z}_Y \simeq \text{or}_X \otimes f^*\text{or}_Y[n - m]$.

(2) Let $D' \subset \mathbb{R}^2$ be locally closed and convex with nonempty interior, and let $j : \text{int } D' \rightarrow D'$ be the inclusion. For every continuous $\Psi : D' \rightarrow \mathbb{R}^2$, $\Psi^!\mathbb{Z} \simeq j_!\mathbb{Z}_{\text{int } D'}$.

Proof. (1) Since ω_Y is invertible, $\mathbb{Z}_Y \simeq \mathbb{D}_Y \omega_Y$, so $f^!\mathbb{Z}_Y \simeq \mathbb{D}_X f^* \omega_Y = \text{RHom}(f^* \text{or}_Y[m], \text{or}_X[n])$. (2) At a non-interior point x , $D' \setminus x$ is star-shaped about an interior point, so the local homology vanishes. Hence $\omega_{D'} \simeq j_!\mathbb{Z}_{\text{int } D'}[2]$, and $\Psi^!\mathbb{Z} \simeq \mathbb{D}_{D'}(\mathbb{Z}_{D'}[2]) = \omega_{D'}[-2]$. $\square$

Corollary 4.5. For Ψ of (3) on $D_\circ$ and any inclusion $i : L \rightarrow D_\circ$, the complex $i^*\Psi^!\mathbb{Z}$ is $\mathbb{Z}$ in degree 0 on $L \cap \text{int } D_\circ$ and 0 on the sides $u = 0, 1$. In particular $i^*\Psi^!\mathbb{Z} \simeq \mathbb{Z}[-1]$ holds at no point, fold or otherwise. For an embedded open arc L in the interior, $i^!\Psi^!\mathbb{Z} \simeq \text{or}_L[-1]$; this shift is the codimension of L , not a feature of folds.

Proposition 4.6 (Where the fold is visible). *Let x be an interior ordinary fold point of Ψ , and choose disc neighbourhoods $V \ni x$ and $W \ni \Psi(x)$ on which $f = \Psi|_V$ is homeomorphic to $(\xi, \eta) \mapsto (\xi, \eta^2)$. Let L be the fold arc, let $W' \subset W$ be the open side of the discriminant containing $f(V \setminus L)$, and put $G = j_! \mathbb{Z}_{W'}$. Then $f^! \mathbb{Z}_W \simeq \mathbb{Z}_V$, $f^* G \simeq (\mathbb{Z}_{V \setminus L})_!$ and $f^! G \simeq \mathbb{Z}_V$. The comparison map $f^! \mathbb{Z}_W \otimes f^* G \rightarrow f^! G$ is injective with cokernel $i_* \mathbb{Z}_L$. Thus the comparison condition of cohomological smoothness fails at every ordinary fold point, while invertibility of $f^! \mathbb{Z}$ holds.*

Proof. For the closed side $k : H \rightarrow W$, local homology vanishes on its edge, so $\mathbb{D}_W(\mathbb{Z}_H) = k_* \omega_H \simeq G[2]$. Since $f(V) \subset H$, $f^! G[2] \simeq \mathbb{D}_V(f^* \mathbb{Z}_H) = \mathbb{Z}_V[2]$. Off L , f is locally an open immersion, so the comparison map is an isomorphism there. A morphism $(\mathbb{Z}_{V \setminus L})_! \rightarrow \mathbb{Z}_V$ is a locally constant function on $V \setminus L$, here with values ± 1 . Its cokernel is $i_* \mathbb{Z}_L$. $\square$

The fold is also visible in $f_* \mathbb{Z}_V$, whose stalk ranks are $2, 1, 0$ across the discriminant, and in the normalization defect on the surface side. A compact Hausdorff image defines a condensed set. That fact alone, however, supplies neither a canonical structure sheaf nor a singularity detector (v12 Remark 8.7).

5 Residual geometry

5.1 Sublevel law and resolution cost

For the standard models take Lebesgue measure and the residuals $K_f = x^2 + y^4$ and $K_\times = x^2(1 + y^2) + y^4$.

Theorem 5.1 (Sublevel law). *With $B = B(\frac{1}{4}, \frac{3}{2})$ and $\varepsilon > 0$, $V_f(\varepsilon) = B\varepsilon^{3/4}$ and $B\varepsilon^{3/4}(1 + \sqrt{\varepsilon})^{-1/2} \leq V_\times(\varepsilon) \leq B\varepsilon^{3/4}$. The exponent $3/4$ survives smooth changes of density and of coordinates; the coefficient does not.*

Proof. Slicing in y gives $4 \int_0^{\varepsilon^{1/4}} \sqrt{\varepsilon - y^4} dy$. Substituting $y = \varepsilon^{1/4} v$ and then $z = v^4$ yields $B\varepsilon^{3/4}$. For the cross-cap, the extra factor $(1 + y^2)^{-1/2}$ lies between $(1 + \sqrt{\varepsilon})^{-1/2}$ and 1. $\square$

The geometric cost $-\log V(\varepsilon) = \frac{3}{4} \log \varepsilon^{-1} - \log B + o(1)$ is a resolution cost, not a channel capacity. Identical binary labels admit a constant stochastic channel of capacity 0 and a noiseless one of capacity $\log 2$. The residual supplies neither the kernel nor the resource constraint. The earlier conjecture $C = \sqrt{5} \ln S$ is withdrawn.

5.2 An explicitly chosen complex completion

Theorem 5.2 (Chosen A_3 completion). *After its coefficients are extended to $\mathbb{C}$, the germ of $K_\times$ is right-equivalent to $X^2 + Y^4$, with Milnor and Tjurina numbers 3. The fiber $U = \{X^2 + Y^4 = 1\}$ is a smooth genus-one curve E with two points removed. $H^1(U)$ has dimension three and monodromy spectrum $\{-i, 1, i\}$. The weight-graded pieces of its mixed Hodge structure have dimension two in weight one and one in weight two.*

Proof. The substitution $X = x\sqrt{1 + y^2}$, $Y = y$ is a holomorphic coordinate change with invertible derivative. The weighted compactification $X^2 + Y^4 = Z^4 \subset \mathbb{P}(2, 1, 1)$ is a double cover of $\mathbb{P}^1$ branched at four points, hence of genus one. Its two ends at infinity are unramified. Weighted homogeneity identifies the global and local Milnor fibers [11, 12]. The monodromy $h(X, Y) = (-X, iY)$ acts by $-i$ and i on $H^1(E)$ and fixes the boundary class. The Gysin sequence $0 \rightarrow H^1(E) \rightarrow H^1(U) \rightarrow \tilde{H}^0(\{\infty_\pm\})(-1) \rightarrow 0$ gives the weights [13, 14]. $\square$

These weights belong to the chosen complexification. They are not inferred from the real surface or from a Mellin parameter. The monodromy alone does not determine the specialization and variation maps of a nearby-cycle package.

5.3 Three different projected curves

Proposition 5.3. (1) For each $r > 0$, the real link $\{K_\times = r^2\}$ is a circle. Its cross-cap image has one transverse double point, so it is a figure-eight graph. The lift (x, xy, y^2, y) is an embedding. (2) The weights of $X^2 + Y^4$ are $(2, 1)$. For $a, b > 0$ with $a^2 + b^4 = 1$, the orbit $\Gamma(s) = (ae^{2is}, be^{is})$ is embedded, and its shadow $(b \sin s, a \sin 2s)$ is a transverse figure-eight with frequency ratio $2 : 1$. The action $2\pi(4a^2 + b^2)$ has infimum 2π over nondegenerate orbits; it is attained only in their closure, and $b^2 = \frac{1}{8}$ is a strict maximum. (3) For the actual lateral germs the Euclidean Whitney invariants satisfy $b_\pm^2 + a_\pm c_\pm^2 = 0$. The exceptional principal-curvature trace has an ordinary cusp at the double return, not a node [15, Prop. 2.1].

Sketch. (1) The gradient of $K_\times$ vanishes only at 0. Equal images force $x = x'$ and $y' = \pm y$, and the tangent images at $(0, \pm\sqrt{r})$ are transverse. (2) The Y coordinate fixes the phase. With $z = b^2$ the action divided by 2π is $1 + (1 - z)(4z + 3)$. (3) The invariants are computed exactly from the actual second jet, and the pedal trace satisfies $\det(p'', p''') \neq 0$ at its double return. The computation is in the exact replay of Appendix B. $\square$

These three constructions yield distinct curves. No amplitude, clock or scattering rule transfers from one to another.

6 Spectral and arithmetic statements that survive

These results are standard. We record them because the earlier versions stated them with gaps in measure, domain or hypotheses.

Mellin operators. Let $T_\sigma f = -i(xf' + \sigma f)$. For C^1 functions on $[a, b] \subset (0, \infty)$, $\int_a^b (\overline{T_\sigma f} g - \overline{f} T_{1-\sigma} g) dx = i[xfg]_a^b$. In $L^2(dx)$ the formal adjoint is therefore $T_{1-\sigma}$, which equals T_σ exactly at $\sigma = \frac{1}{2}$. For the measure $x^{2\beta-1} dx$ the formal adjoint parameter is $2\beta - \sigma$ instead. Formal symmetry is not self-adjointness. The map $(Uf)(s) = e^{s/2} f(e^s)$ is unitary from $L^2([1, \infty), dx)$ onto $L^2([0, \infty), ds)$, and it carries the closure of $T_{1/2}$ on C_c^∞ to $-i\partial_s$ on H_0^1 , whose adjoint has domain H^1 . The deficiency indices are $(1, 0)$, so no self-adjoint extension exists. On the full multiplicative line the operator is self-adjoint [16]. The functions $x^{-\sigma-i\omega}$ satisfy $T_\sigma x^{-\sigma-i\omega} = -\omega x^{-\sigma-i\omega}$. They lie in $L^2([1, \infty))$ exactly when $\sigma > \frac{1}{2}$, and even then they violate the minimal zero trace. Real formal eigenvalues imply no orthogonality.

Finite Gram matrices. For distinct real $\omega_1, \dots, \omega_M$, $T > 0$ and $G_{jk} = T^{-1} \int_0^T e^{i(\omega_j - \omega_k)t} dt$, the phasors are linearly independent and G is positive definite for every $T > 0$; a Vandermonde argument at an interior time proves this. With $\varepsilon = 2(M-1)/(T \min |\omega_j - \omega_k|) < 1$, the spectrum of G lies in $[1 - \varepsilon, 1 + \varepsilon]$, $\kappa_2(G) \leq (1 + \varepsilon)/(1 - \varepsilon)$ and $M \log(1 - \varepsilon) \leq \log \det G \leq 0$. The earlier threshold $T > 2(M-1)/\delta$ is thus sufficient for conditioning but not needed for independence. These estimates concern a fixed finite family.

Hardy evaluation and Bohr–Haar. On the coefficient Hilbert space of Dirichlet series, evaluation at $\Re s = \sigma$ is bounded exactly when $\sigma > \frac{1}{2}$, with norm $\zeta(2\sigma)^{1/2}$ [17]. The Gram limit $\lim T^{-1} \int_0^T n^{it} m^{-it} dt = \delta_{nm}$ is character orthogonality on the product torus of prime exponents. A finite set of integers may generate a lattice of lower rank, as $\{2, 4\}$ does, so no free group of rank M is involved. The half-density threshold and the evaluation threshold are two separate results.

Proposition 6.1 (Log-scale and linear-scale windows). *Let $K(\delta, L) = L^{-1} \int_0^L e^{-i\delta\tau} d\tau$, so that $|K| = |\sin(\delta L/2)|/(\delta|L/2|) \leq 2/(\delta|L|)$, and let $T > 1$.*

1. $\frac{1}{\log T} \int_1^T x^{-i\delta} \frac{dx}{x} = K(\delta, \log T)$. The log-scale and linear-scale bounds are one bound, $2/(|\delta| \cdot \text{mass})$, evaluated at window masses $\log T$ and T ; their ratio $T/\log T$ is the same for every real δ .
2. For the same characters on the same interval with Lebesgue measure, $\frac{1}{T-1} \int_1^T x^{-i\delta} dx = \frac{T^{1-i\delta}-1}{(1-i\delta)(T-1)}$. Its modulus tends to $(1+\delta^2)^{-1/2} \neq 0$, so these characters do not decorrelate at all.
3. For $e^{-i\delta t}$ on $[1, T]$ with Haar measure the average is at most $2/(|\delta| \log T)$ in modulus.

Proof. Substitute $x = e^\tau$; integrate directly; integrate by parts once. $\square$

No integrality enters any of these statements. Moreover $\mathbb{N}$ is not a subgroup of $(\mathbb{R}_{>0}, \times)$; the subgroup it generates, $\mathbb{Q}_{>0}$, is dense. The “arithmetic decorrelation gap” of the earlier public description therefore has nothing to explain.

7 Scope limits

The following limits were already stated correctly in v12. We retain them:

- A nonempty proper subset of $SU(2)$ is not invariant under the full left action. This applies to a constrained source lift once one is specified.
- The ratio $2\pi^2 R^3/(\pi r^2)$ with $r = \phi^{-2}R$ equals $2\pi\phi^4 R$. It carries a length scale and defines no mass operator.
- For the canonical Kähler form the two power coordinates Poisson-commute, whereas the full $SU(2)$ moment map is a different observable algebra.
- The circle operator $-i\partial_\theta$ with antiperiodic condition has spectrum $\mathbb{Z} + \frac{1}{2}$, which is symmetric, so its eta invariant vanishes [18].
- The real geometry specifies no complex algebraic pair, finite-field model or Frobenius. A Weil cohomology theory is extra input. Given such a theory with localization and Tate twists, the pair (E, D) of Theorem 5.2 has a one-dimensional boundary term.

8 What remains open

- Whether a quadratic readout realizes S on D_+ under the historical powers (Remark 3.10). Whether any principle selects a canonical non-analytic readout; Theorem 3.8 suggests that none does without further input.
- A sign-twisted, form or spinor operator whose boundary data are compatible with the carriers of Proposition 4.3.
- The actual singular Dirac and Pin problem of v12 Sections 8.20–8.24: closed domains for the twisted Dirac operator on the actual Whitney metric, graph and overlap estimates, singular-cut trace spaces, and Clifford-compatible transmission maps. Exact-cone and regular-cut results remain model or conditional statements. The scalar theory needed as a first step is developed in [4].
- Independent signal or physical models for capacity, mass, clocks and chirality. None follows from the geometry above.

A Corrections to the earlier record

A.1 The public description

The description field of the record [1] still shows the abstract of an earlier draft, which differs from both public files. Its claims are corrected as follows:

Claim in the description	Status here
S is derived from the moment map and the power constraint	False for analytic readouts (Theorem 3.6); smooth realizations exist but are not canonical
Immersion singularities occur at exactly two lateral points	Five rank labels (Proposition 2.1)
Quasi-separated condensed set; Weil-étale/Pontryagin connection	No such structure or bridge (§6, Proposition 4.3(1), v12 7.7, 8.5, 8.7)
$i^*\Psi^1\mathbb{Z} \simeq \mathbb{Z}[-1]$ on the fold locus	False at every point (Corollary 4.5)
Monodromy as an endomorphism of the constant condensed sheaf	$\text{End}(\mathbb{Z}) = \mathbb{Z}$; carrier is $\pi_*\mathbb{Z}$ (Proposition 4.3)
$T > 2(M-1)/\delta$ gives independence	Independence holds for all $T > 0$ (§6)
$C = \sqrt{5} \ln S$ (conjectural)	Withdrawn; resolution law and capacity no-go (§5)

A.2 v11 Theorem 4.2

v11 identified the sign reversal of $\sqrt{F}$ with an exchange of the sheets of a Whitney fold. v12 already withdrew that identification; Proposition 4.3(3) shows the two involutions are different. In all other blocks up to 8.20, v11 agrees with v12, apart from small wording refinements. v11 stated the spinorial parametrix only as a remark, more cautiously than v12.

A.3 All named statements of v12

Categories: **H** holds, possibly with an explicit domain or hypothesis; **C** corrected statement; **S** strengthened; **R** refuted as defined; **O** unproved, now open; **M** model-level or conditional only; **D** definition gap, decided here. “Companion” refers to [4].

Block	Cat.	Note and location
3.1	D	No analytic readout under historical powers; smooth readout exists; other powers admit a quadratic readout (§3)
3.2	C	Fix $L^2(dx)$ and the conjugate-linear first slot; the introduction used another measure (§6)
3.3	H	Geometric resolution cost, not a capacity (§5)
4.1	C	F is an angular-chart field on D_0 ; indices ∓ 1 hold (§2, Thm 4.1)
4.2	H	Two-traversal sign restoration, no clock (Thm 4.1)
4.3	H	No spin identification is claimed
4.4	C	N and M are both even; the parity explanation is replaced (§4)
5.1	H	$ P'(c_0) = \sqrt{5}$ for the monic P only
5.2	C	The lateral Jacobian modulus is not $\sqrt{5}$; its sign follows the orientation
5.3	H	Exact fold constant; two-sided cross-cap bounds (Thm 5.1)
5.4	H	Requires an external identification $S = \varepsilon^{-1}$
5.5	H	Capacity no-go
5.6	H	Explicit holomorphic coordinate change (Thm 5.2)
5.7	H	Genus one, two unramified ends, spectrum $\{-i, 1, i\}$
5.8	H	Every $r > 0$ for the standard model (Prop. 5.3)
5.9	H	The orbit leaves the fixed fiber except at fourth-root phases
5.10	C	The minimum is attained only in the closure; $b^2 = \frac{1}{8}$ is a maximum
5.11	H	Scope remark
5.12	H	Invariants rebuilt; ordinary cusp criterion
5.13	H	Geometric no-go

Block	Cat.	Note and location
6.1	C	$c^2 - 3c + 1$ selects only the lateral labels; five rank labels (Prop. 2.1)
6.2	C	Five cross-cap germs, including two corner quarter-germs
7.1	C	The formal adjoint parameter depends on the measure; formal symmetry is not adjoint-domain equality
7.2	H	Indices $(1, 0)$ with explicit H_0^1 and H^1 domains
7.3	C	For $\sigma > \frac{1}{2}$ the functions are L^2 but violate the zero trace
7.4	H	Conventions made explicit
7.5	H	For a fixed finite family
7.6	S	Independence holds for every $T > 0$; the threshold only controls conditioning
7.7	H	Bohr torus realization
7.8	H	Conditioning and log-determinant bounds
8.1	H	Full-line and half-line realizations are distinguished
8.2	H	Evaluation boundary $\sigma = \frac{1}{2}$
8.3	H	The coincidence of two thresholds carries no weight interpretation
8.4	H	Character orthogonality
8.5	H	Lattices of lower rank are retained
8.6	H	Normalization defect (Prop. 4.3(4))
8.7	H	Strengthened by Thm 4.4
8.8	C	The displayed remainder omitted a term; the fold holds through exact charts (Prop. 4.2)
8.9	C	All five labels must be removed; $H_1 = \mathbb{Z}^2$
8.10	H	Exact Galerkin pullback (Companion)
8.11	C	Zero capacity at all five labels; form closures, not graph closures (Companion)
8.12	O	Model identities hold; the actual uniform estimate is open (Companion)
8.13	R	The graph closure of the interior core has infinite outer-boundary deficiency; the corrected two-point operator is treated in the Companion
8.14	R	False for the old minimum; holds for the corrected operator (Companion)
8.15	R	The Krein kernel of the old minimum is infinite; compactness and gap hold (Companion)
8.16	H	Buckling equivalence for the actual operator, with hypotheses proved (Companion)
8.17	C	Operator scope corrected (Companion)
8.18	H	Support and parity obstruction (Prop. 4.3)
8.19	H	Scope remark
8.20	M	Exact-cone model only (§8)
8.21	O	Actual front-seam parametrix unproved (§8)
8.22	M	First-order data hold; a Pin convention must be specified
8.23	M	Conditional on an unspecified Clifford-compatible transmission
8.24	H	Open-problem statement
8.25	H	Conditional on a specified proper lift (§7)
8.26	H	Representation observation only
8.27	H	Real-geometry Deligne/Weil no-go
8.28	H	Weights of the chosen completion (Thm 5.2)
8.29	H	Monodromy does not fix the full nearby-cycle package
8.30	H	Conditional on a specified Weil realization
9.1	H	The ratio carries a length scale
9.2	H	No mass prediction
9.3	H	Power coordinates Poisson-commute
9.4	H	A separate observable programme
9.5	H	$\eta = 0$ for the sign circle
9.6	H	No chirality from the sign cover

B Evidence and reproducibility

The historical authority is the public v12 file, which is byte-identical to the archived copy. The SHA-256 values of the two public files are

```
v12  4ebb8f0a7d6e88a6c5cd97f65d224b4e938867cabad54f057d3ceafb24167d5a
v11  207a51feed62fe7f5a6dd7c9190258dd1b9c996c8b2735e799aff569bb017ec2
```

Exact and finite replay. Independent scripts check the following:

- derivative determinants, the Euclidean jet invariants and the polar remainder;
- the quadrature identities and the Bloch identity;
- the edge and skew-line algebra;
- a replay of the C^∞ readout on 64,881 labels;
- the analytic edge-chart identities on the half source;
- the Jacobian factorization, the windings ∓ 1 and the idempotents of $\mathbb{Z}[C_2]$;
- the window kernels.

None of these replays certifies an analytic, sheaf-theoretic or operator-domain statement.

Lean. A companion Lean project (Lean 4, Mathlib) contains 33 named theorems, audited for proof holes and custom axioms. They include the actual rank classification, the endpoint ranks, the polar fold, the small-circle indices, the evenness of the field, the monic derivative, the injectivity and fibres of the cross-cap lift, the square-root and logarithm lifting, the Mellin interval identity, the Gram entries and eigenvalue enclosure, the quadrature readout identities and the cusp factorization. This is partial coverage of selected statements, not a formalization of the paper.

External theorems. The paper relies on the following external results, each with its hypotheses verified for the actual maps:

- smooth cross-cap recognition [5];
- Milnor fiber comparison and mixed Hodge theory [11, 12, 13, 14];
- the Euclidean pedal formula [15];
- Verdier duality and six-functor smoothness [8, 7, 9];
- Dirichlet–Hardy space facts [17];
- self-adjointness by Fourier transform [16].

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